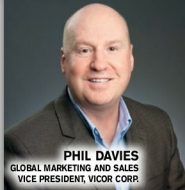


Power System Design:

Whiteboard... would give them time to work on multiple projects

Power system design advancements have provided power engineers with many tools and technologies. Let us take a look at what some of these are. Phil Davies, global marketing and sales VP, Vicor Corp., speaks with Atul Goel and Dilin Anand from EFY



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Q. What is meant by power component methodology?

A. It is for almost all power control and regulation engineers. Any engineering work being done on AC-DC or DC-AC at any power level can be considered to be a part of it, including front-end power systems developing voltage to be used as a hub from 48V (or, 28V in military) to the point of low voltages. The methodology aims to make it simple, scalable, repeatable, deliver faster time-to-market and for a design manager to get more productivity out of his design engineers.

Q. How do power regulation components help engineers?

A. These components typically switch at a very high frequency, which enables reduction in the size of magnetics and all types of passives that go within it. Having a lot of intellectual property in planar magnetics gives engineers the ability to put cores of transformers and inductors inside the packages.

In our case, this, coupled with proprietary conversion and regulation engines, and the fact that we have our own silicon team, allows us to design control integrated circuits (ICs) ourselves. We leverage this with our packaging technology to build the densest and most efficient power conversion and regulation modules.

Q. Please elaborate on these engines?

A. A basic cornerstone of these engines is the sine amplitude converter, which is similar to a resonant converter. When we use this in bus converter modules and voltage multipliers, there is no energy storage. You can pulse very high current through here, with very fast transient response.

The ZVS zero-voltage switching topology engine allows switching at very high frequencies, but when switched on or in transition, it is done at zero voltage and, in some cases, zero current, which minimises switching losses and body recovery all through the diodes in the power metal oxide semiconductor field effect transistors and other devices.

Q. What technological improvements have helped these components in the past few years?

A. Improvements over the years have been with respect to switching at higher frequencies, as well as with integrated core planar magnetics technology.

Earlier, these chips were manufactured single-sided, but are now being manufactured double-sided. This lets designers ensure that heat is drawn out of the component from both sides of the package. By extracting the heat faster, they

get better watts per cubic centimetre or watts per square inch, which further reduces the density of the product.

Moulding material, PCBs and silicon integration are other areas we focus on. We now integrate more, moving from Bipolar to BiCMOS to Bipolar-CMOS-DMOS technology.

Q. Has the design process of a power system changed?

A. Many a time, a company develops a new electronics system and then goes to the engineer with the requirements. They are often considered right at a later stage and end up designing their own power converters over and over.

Q. Could they not re-use a design?

A. It is very difficult to leverage one design from one platform to the next. Our methodology suggests using power modules from front-end to the point of load and use these to rapidly design a power system instead of starting from scratch. This allows engineers to increase productivity with respect to engineering time, enabling them to work on more designs. This way they can leverage the same components again on the next system design.

Q. How is designing power systems being made easier?

A. Whiteboard is a tool that lets you sit with your colleagues to architect a power system without being in the same geographical area. It supports dragging and dropping components like a front-end AC power supply, specifying the input and output voltage or building an entire system out of different chips and building blocks. Then it shows the efficiency of the system and the actual mechanical picture of the system, so that you can understand the density and sizes of different components. Then, if you want to do a simulation of the design, you can do that by looking at transient response, output ripple or change-switching frequencies.

Q. Could these design tools help non-experts?

A. Yes, they will be able to quickly build designs by using the whiteboard to simulate the solution. They can then use the evaluation boards to test it very quickly.

Q. How do engineers handle constant simplification of the design process?

A. Some engineers might feel this could replace their skills, but at the same time, some might feel that this would give them time to work on multiple projects, faster. ●